Finance Tracker - Stateful Cloud-Native Application

ECE1779 Group 19

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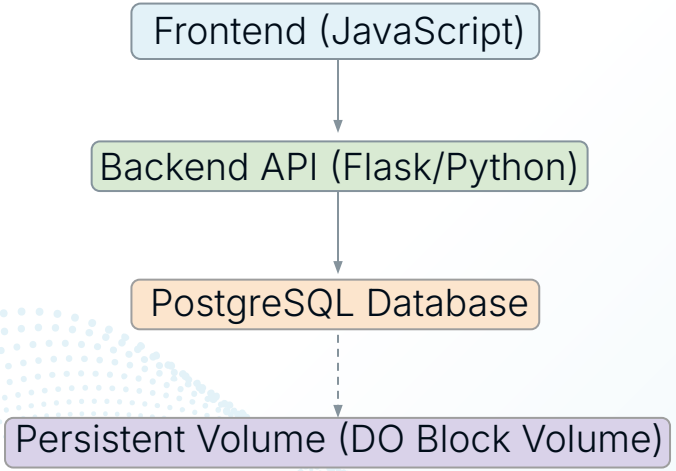
Introductions

- Centralized financial tracking application
- Persistent storage
- Accessible from anywhere
- Supports:
  - Accounts
  - Transactions
  - Balances

Demo

Architecture



Orchestrated with Kubernetes (DOKS)

Core requirements

Requirement	Implementation
Docker containerization	Docker + Docker Compose
PostgreSQL with persistent storage	DigitalOcean Volume
Orchestration	Kubernetes through DOKS
Monitoring and observability	Prometheus + Grafana + DigitalOcean Alerts
Multi-container setup	Frontend + Backend + Database
Stateful Design	Persistent Database

Persistence & Stateful Design

- PostgreSQL with:
  - WAL
  - Fsync enabled
- Persistent Volume (Block storage)
- Data survives:
  - Container restart
  - Pod failure

Kubernetes Deployment

- Backend deployment with 2 replicas
- Frontend deployment
- StatefulSet database
- PersistentVolumeClaim (PVC)
- Ingress for public access

Monitoring & Observability

- Primary Observability Stack in Kubernetes:
  - **Node Exporter + Kubelet** → infrastructure metrics
  - **Flask /metrics endpoint** → application metrics
  - **Prometheus** → collecting metrics & storing in its persistence storage
  - **Grafana** → visualization & dashboards
- Additional Application-Level Observability:
  - Health endpoints
    - /health → app status
    - /health/db → DB connectivity check
  - Lightweight request/response logging
- Platform-Level monitoring via DigitalOcean
  - Dashboards (CPU, memory, disk, bandwidth, etc.)
  - Sending alerts when resource utilization > 80%

Grafana dashboards demonstrating live application metrics

Grafana dashboards demonstrating live resource usage

Advanced Features

- Backup & Recovery
  - Daily dump
  - Stored in DO spaces
  - Restore capability
- Security Enhancements
  - JWT authentication
  - Data hashing
  - Data isolation

Future Plan

- Advanced features implementation
  - ➢ CI/CD Pipeline or WebSockets implementation
- Further polishing
  - ➢ Monitoring (metrics/alerts) dashboard on UI

Thanks for listening!